

Radio network

Problem ID : problem-36-22913428

Time limit : 3s

Telstra plans to build a new radio network connecting n stations across Perth. There are m possible fibre links between pairs of stations. Each link has a construction cost between 1 and c . Telstra wants all stations to be connected, directly or indirectly, while minimising the total construction cost.

Determine the smallest total construction cost that allows messages to travel between any pair of stations in the finished network.

Input

The first line contains three integers n , m and C , ($1 \leq n \leq 10^5$, $0 \leq m \leq 10^5$, $1 \leq C \leq 100$) representing the number of stations, the number of possible fibre links and the maximum construction cost.

Each of the next m lines contains three integers u , v and w , ($1 \leq u, v \leq n$, $u \neq v$, $1 \leq w \leq C$) representing a possible fibre link between stations u and v with construction cost w . The stations are numbered from 1 to n . Each fibre link allows communication in both directions. Multiple fibre links may exist between the same pair of stations.

Output

Output the minimum total construction cost that allows messages to travel between every pair of stations. If this is impossible, output -1.

Sample input 1

```
5 7 5
1 2 3
1 3 1
2 3 2
2 4 4
3 4 3
3 5 5
4 5 2
```

Sample output 1

```
8
```

Sample input 2

```
4 2 5
1 2 1
3 4 2
```

Sample output 2

```
-1
```